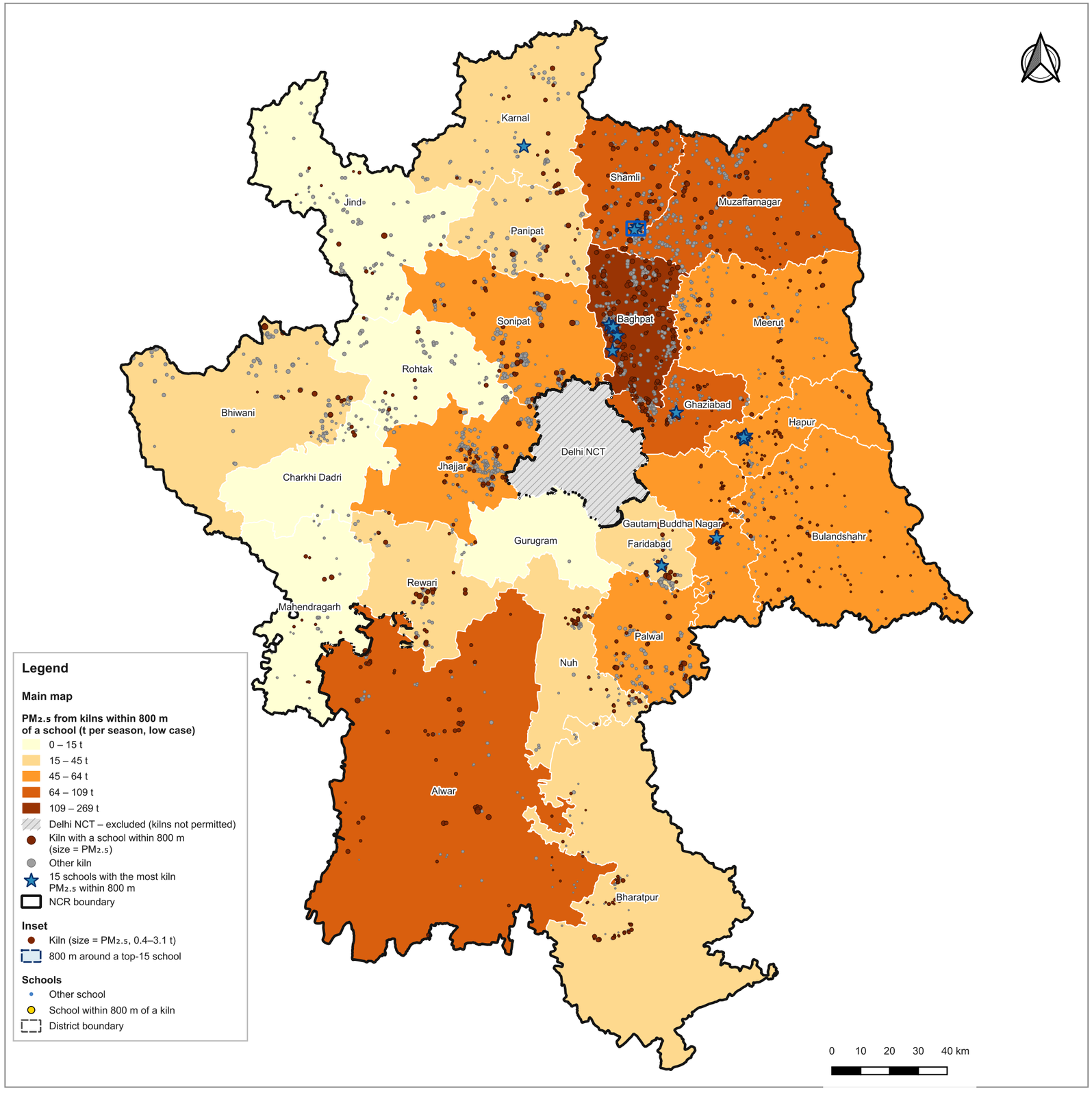
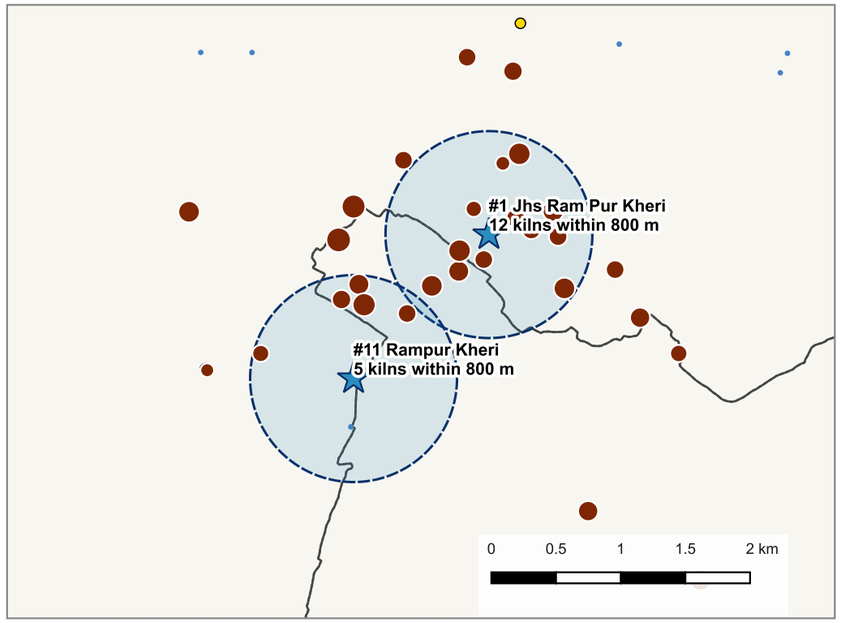
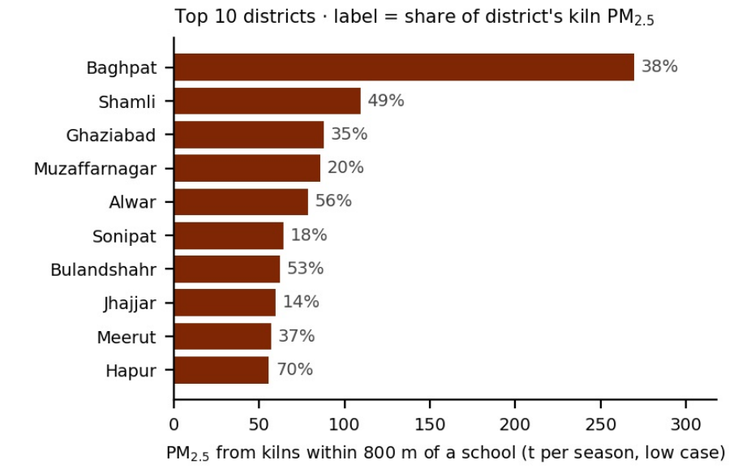
A third of brick-kiln smoke in the NCR is released within 800 m of a school

Estimated $PM_{2.5}$ emitted by brick kilns in one firing season, and the share from kilns within the CPCB 800 m siting distance of a school · National Capital Region, 2023–24





Inset: Rampur Kheri, Shamli – most kilns within 800 m of a school (blue box on main map)



Key numbers (NCR excluding Delhi)

≈ 4,400–11,600 t $PM_{2.5}$ per season from 3,057 kilns (low: zigzag, 4 M bricks/kiln; high: fixed chimney, 6 M)
30% of it (1,300–3,500 t) comes from the 33% of kilns that have a school within 800 m
Baghpat: 270 t from kilns near schools (low case), 2.5× the next district
Most-affected schools (kilns within 800 m · t $PM_{2.5}$, low case):
1 JHS Rampur Kheri, Shamli – 12 · 25 t 2 P.S. Matiyala, Ghaziabad – 10 · 15 t
3 UPS Biharipur, Baghpat – 7 · 15 t 4 N.K. Memorial Public School, Dahapa, Hapur – 8 · 14 t
Emissions at source, not air quality at the school (no dispersion model).

Method: $PM_{2.5}$ = bricks × 3 kg × emission factor. Emission factors: World Bank (2020) Table 2.1 (zigzag 0.12, fixed chimney 0.21 g/kg fired brick). Bricks per kiln ASSUMED (4–6 million/season). Each kiln's share set by its 2023–24 Sentinel-2 firing frequency. Kilns: YOLO11 detector, firing-verified, non-kilns removed, every kiln checked by eye (doubtful ones by R. Sinha). Schools: UDISE (DataMeet), location-checked, shared points removed.
Data: Copernicus Sentinel-2 (ESA), SentinelKilnDB (IIT Gandhinagar, CC BY-NC 4.0), ESA WorldCover, UDISE/DataMeet, World Bank (CC BY 3.0 IGO), OpenStreetMap contributors, NCRPB boundaries (rebuilt).
CRS: WGS 84 / UTM zone 43N (EPSG:32643) · Produced 16 October 2026 · KilnWatch – Rashmi Sinha